Homework 8

Graded

Student

Colin Skinner

Total Points

89.5 / 100 pts

Question 1

Problem 1 — **19** / 20 pts

1.1 a — **9** / 10 pts

- **– 0 pts** Correct
- **– 4 pts** Major Error/Partially Incomplete
- ✔ **– 1 pt** Minor Error
- **– 10 pts** Incomplete

1.2 b — **10** / 10 pts

- ✔ **– 0 pts** Correct
- **– 4 pts** Major Error/Incomplete
- **– 2 pts** Minor Error
- **– 10 pts** Incomplete

Question 2

Problem 2 — **8** / 10 pts

- **– 0 pts** Correct
- ✔ **– 2 pts** Minor error
- **– 10 pts** Incomplete
- **– 1 pt** Sign error
- **– 0 pts** Click here to replace this description.
- **– 5 pts** Major Error/Incomplete

Question 3

Problem 3

10 / 15 pts

- **– 0 pts** Correct
- **– 3 pts** Small errors in theta dot derivation or incomplete expression
- **– 15 pts** Incomplete
- **– 5 pts** Does not solve for terms / no conservation
- ✔ **– 5 pts** Major Errors

Question 4

Problem 4 — **19** / 20 pts

4.1 **a** — **19** / 20 pts

- **– 0 pts** Correct
- ✔ **– 1 pt** Minor Error a)
- **– 1 pt** Minor Error b)
- **– 1 pt** Minor Error c)
- **– 20 pts** no attempt
- **– 5 pts** No attempt at b

Question 5

Problem 5 — **13.5** / 20 pts

5.1 **a** — **6.5** / 10 pts

- **– 0 pts** Correct
- ✔ **– 3.5 pts** Correct approach, significant errors
- **– 6 pts** Major Errors
- **– 10 pts** Incomplete

5.2 **b** — **7** / 10 pts

- **– 0 pts** Correct
- **– 1.5 pts** Minor error/carryover error
- ✔ **– 3 pts** Significant error
- **– 10 pts** Incomplete

Question 6

Problem 6 **20** / 15 pts

6.1 **Main** **15** / 15 pts

✔ **– 0 pts** Correct

– 15 pts no sumbission

6.2 **Extra Credit** **5** / 0 pts

✔ **+ 5 pts** Correct

+ 0 pts Click here to replace this description.

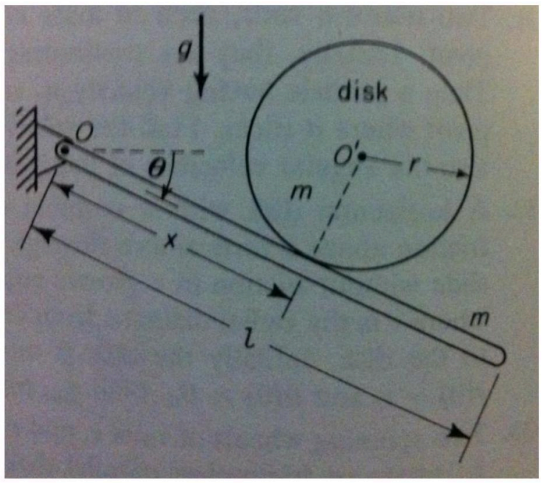
No questions assigned to the following page.

AA 242A Homework 8

Assigned: **Wednesday, November 19th, 2025**

Due: **Friday, December 3rd, 2025 at 11:59 PM**

1. Greenwood 7-36

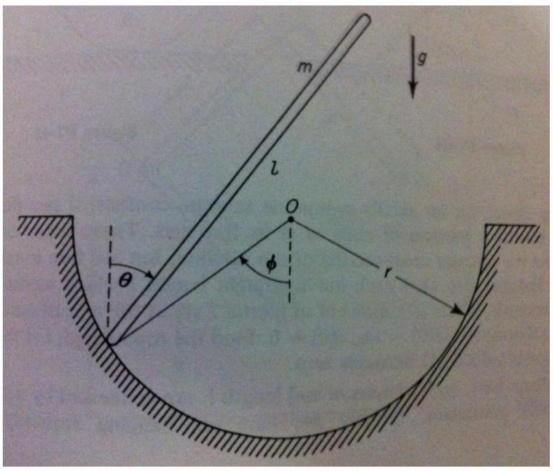


A disk of mass m and radius r can roll without slipping on a rod of mass m and length $l = 4r$ which is pivoted at one end.

a.) Using x and θ as generalized coordinates, obtain the differential equations of motion.

b.) Assuming that the system is released from rest with $\theta(0) = 0$, $x(0) = 2r$, solve for the initial force of the disk acting on the rod.

2. Greenwood 7-41

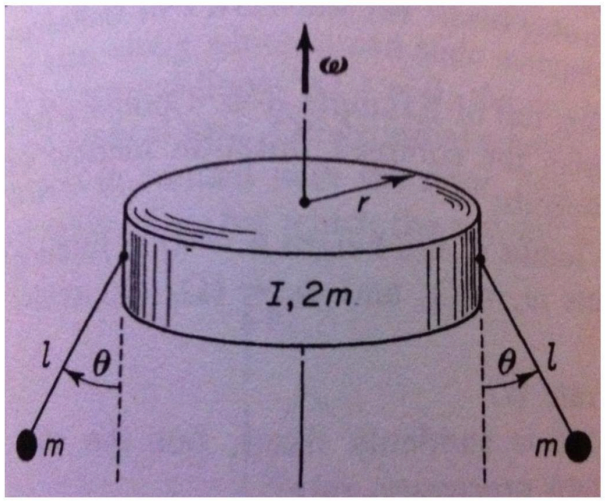


A thin uniform rod of mass m and length l undergoes planar motion on the smooth surface of a fixed hemispherical bowl of radius r. Using θ and ϕ as generalized coordinates, obtain the differential equations of motion.

No questions assigned to the following page.

A thin uniform rod of mass m and length l undergoes planar motion on the smooth surface of a fixed hemispherical bowl of radius r. Using θ and ϕ as generalized coordinates, obtain the differential equations of motion.

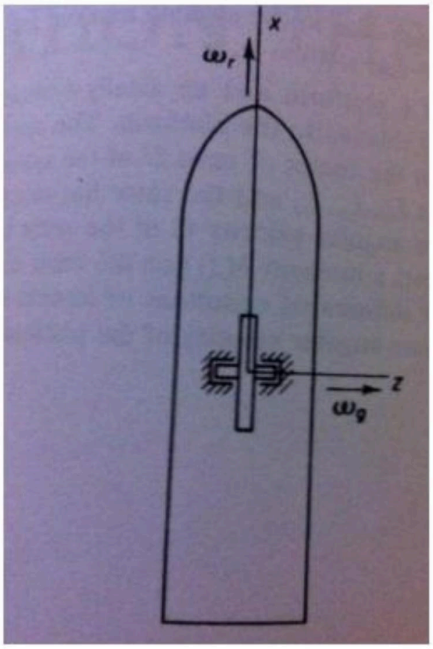
3. Greenwood 8-14



A satellite consists of a disk or radius r, mass $2m$, and moment of inertia I about the axis of symmetry. Two particles, each of mass m, are attached by rigid massless rods of length l to opposite points on the circumference. The frictionless pin joints are oriented so that the two rods always lie in a plane which contains the symmetry axis and rotates with the disk. Initially, the disk is spinning about its axis of symmetry at ω_0 rad/sec with $\theta(0) = 0$ and $\dot{\theta}(0) = 0$. For free motion, solve for the satellite rotation rate ω and also the value of $\dot{\theta}$ as functions of θ.

No questions assigned to the following page.

4. Greenwood 8-34



An axially symmetric rocket is spinning in free space about its axis of symmetry with an angular velocity $\omega_r = 10$ rad/sec. Freely suspended at the mass center of the rocket by massless gimbals is a gyro rotor which is rotating at an absolute rate $\omega_g = 10^4$ rad/sec about its own axis of symmetry which is perpendicular to that of the rocket. Suddenly the gimbals are clamped and the rotor motion relative to the rocket is stopped. The gyro rotor has moments of inertia $I_a = 0.1$ kg·m^2 and $I_t = 0.05$ kg·m^2. The rocket, exclusive of the rotor, has moments of inertia $I_a = 10^3$ kg·m^2 and $I_t = 10^4$ kg·m^2. Solve for the following:

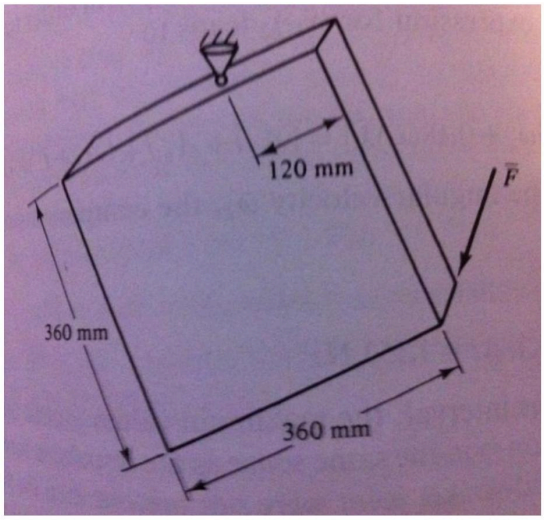
a.) The *x, y, z* components of the angular impulse applied to the rocket by the gyro at the instant the rotor is stopped

b.) The maximum angular deviation of the axis of symmetry of the rocket from its initial position. Note that approximate axial symmetry is maintained after the gimbals are clamped.

c.) Find the time required for a complete precession cycle of the rocket.

No questions assigned to the following page.

5.



A 10-kg square plate suspended by a ball-and-socket joint is at rest when it is struck by a hammer. The impulse force $\vec{F}$ generated by the hammer is normal to the surface of the plate, and its average value during the 4-ms interval that it acts is 5,000 N. Find:

a.) The angular velocity of the plate at the instant following the impact.

b.) The average reaction at the support.

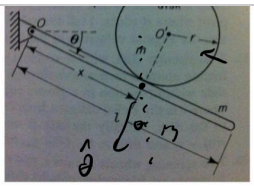
6. For this last week on your creative dynamics problem, solve the problem using Force Balance OR Lagrange. Don't forget to draw your FBD and provide a short description of what you are trying to solve. Also please label your reference frame(s) and coordinate system(s) on your drawing.

   Also, as an exercise, copy and paste your creative problem prompt into ChatGPT, and share the response. Make some comments about what it did right and what it did wrong.

   +5pts extra credit: Solve using both Force Balance AND Lagrange, and compare your answers. If they differ, try to explain why (i.e. different coordinate systems, Lagrange doesn't handle variable-mass systems well, etc.)

   Hint: use Euler angles as your generalized coordinates for 3D rigid body rotation.

Question assigned to the following page: 1.1



A disk of mass m and radius r can roll without slipping on a rod of mass m and length $l = 4r$ which is pivoted at one end.

a.) Using x and θ as generalized coordinates, obtain the differential equations of motion.

b.) Assuming that the system is released from rest with $\theta(0) = 0$, $x(0) = 2r$, solve for the initial force of the disk acting on the rod.

coords: θ, x $\quad \vec{r}_o = x\hat{r} - r\hat{\theta}$

$\vec{v}_o = \dot{x}\hat{r} + x\dot{\theta}\hat{\theta} - r\dot{\theta}\hat{r} = \dot{x}\hat{r} + x\dot{\theta}\hat{\theta} + r\dot{\theta}\hat{r}$

$= (\dot{x} + r\dot{\theta})\hat{r} + x\dot{\theta}\hat{\theta}$

$T_d = \frac{1}{2}mv^2 = \frac{1}{2}m\left[(\dot{x}+r\dot{\theta})^2 + x^2\dot{\theta}^2\right]$

$\dot{x} = r\dot{\theta}_d \qquad T_{rot} = \frac{1}{2}\dot{\theta}_d^2 I = \frac{1}{2}\frac{\dot{x}^2}{r^2}\frac{1}{2}mr^2 = \frac{1}{4}m\dot{x}^2$

$T_{rod} = \frac{1}{2}\dot{\theta}_r^2 I = \frac{1}{2}\dot{\theta}^2\frac{1}{3}m(4r)^2 = \frac{8}{3}mr^2\dot{\theta}^2$

$V = V_r + V_d = mg(-2r\sin\theta) + mg(-x\sin\theta + r\cos\theta)$

$= -2mgr\sin\theta - mgx\sin\theta + mgr\cos\theta$

$\mathcal{L} = \frac{1}{2}m\left[(\dot{x}+r\dot{\theta})^2 + x^2\dot{\theta}^2\right] + \frac{1}{4}m\dot{x}^2 + \frac{8}{3}mr^2\dot{\theta}^2 + 2mgr\sin\theta + mgx\sin\theta - mgr\cos\theta$

$\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{1}{2}m\left(2(\dot{x}+r\dot{\theta})\right) + m\dot{x} = m(\dot{x}+r\dot{\theta}) + \frac{1}{2}m\dot{x} = \frac{3}{2}m\dot{x} + mr\dot{\theta} \qquad \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{3}{2}m\ddot{x} + mr\ddot{\theta}$

$\frac{\partial \mathcal{L}}{\partial x} = \frac{1}{2}m\left(2x\dot{\theta}^2\right) + mg\sin\theta$

$\frac{3}{2}\ddot{x} + r\ddot{\theta} - x\dot{\theta}^2 - g\sin\theta = 0$

$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m(\dot{x}+r\dot{\theta})r + mx^2\dot{\theta} + \frac{16}{3}mr^2\dot{\theta} = m\left[r\dot{x} + \left(r^2 + x^2 + \frac{16}{3}r^2\right)\dot{\theta}\right]$

$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m\left[r\ddot{x} + 2m\dot{x}x\dot{\theta} + \left(r^2 + \dot{x} + \frac{16}{3}mr^2\right)\ddot{\theta}\right]$

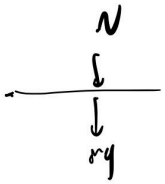
$\frac{\partial \mathcal{L}}{\partial \theta} = 2mgr\cos\theta + mgx\cos\theta + mgr\sin\theta$

$r\ddot{x} + 2m\dot{x}x\dot{\theta} + \left(x^2 + \frac{19}{3}r^2\right)\ddot{\theta} - g(2r\cos\theta + x\cos\theta + r\sin\theta) = 0$

Question assigned to the following page: 1.2

disk:  $N - mg = m_d a_d$

$a_d = \frac{N}{m} - g$

rod: 

$I\alpha = -mg(2r) - N(2r)$

$I\frac{a_r}{2r} = -mg(2r) - N(2r)$

$I_{rod} a_r = -4mgr^2 - 4Nr^2$

$I_{rod}\left(\frac{N}{m} - g\right) = -4mgr^2 - 4Nr^2$

$I_r N - I_r m g = -4m^2 g r^2 - 4mNr^2$

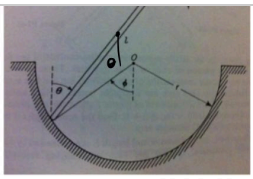
$N(I + 4mr^2) = I mg - 4m^2 gr^2$

$I_{rod} = \frac{1}{3} m \overset{4r}{\downarrow} \ell^2 = \frac{16}{3} mr^2$

$N = \frac{I mg - 4m^2 gr^2}{I + 4mr^2} = \frac{\frac{16}{3} m^2 r^2 g - 4m^2 gr^2}{\frac{16}{3} mr^2 + 4mr^2} = \frac{16 m^2 r^2 g - 12 m^2 r^2 g}{16 mr^2 + 12 mr^2}$

$N = \frac{4}{28} mg$

Question assigned to the following page: 2



A thin uniform rod of mass m and length l undergoes planar motion on the smooth surface of a fixed hemispherical bowl of radius r. Using θ and ϕ as generalized coordinates, obtain the differential equations of motion.

$$\vec{r}_c^{\,s} = -r\sin\phi\,\hat{i} - r\cos\phi\,\hat{j}$$

$$\hat{y}' = \sin\theta\,\hat{i} + \cos\theta\,\hat{j}$$

$$\vec{r}_{cm} = \vec{r}_c^{\,s} + \frac{l}{2}\hat{y}' = \left(-r\sin\phi + \frac{l}{2}\sin\theta\right)\hat{i} + \left(-r\cos\phi + \frac{l}{2}\cos\theta\right)\hat{j}$$

$$\vec{v}_{cm} = \left(-r\cos\phi\,\dot{\phi} + \frac{l}{2}\cos\theta\,\dot{\theta}\right)\hat{i} + \left(r\sin\phi\,\dot{\phi} - \frac{l}{2}\sin\theta\,\dot{\theta}\right)\hat{j}$$

$$v_{cm}^2 = \left(-r\dot{\phi}\cos\phi + \frac{l}{2}\dot{\theta}\cos\theta\right)^2 + \left(r\dot{\phi}\sin\phi - \frac{l}{2}\dot{\theta}\sin\theta\right)^2$$

$$= r^2\dot{\phi}^2\cos^2\phi - rl\dot{\phi}\dot{\theta}\cos\phi\cos\theta + \frac{l^2}{4}\dot{\theta}^2\cos^2\theta + r^2\dot{\phi}^2\sin^2\phi - rl\dot{\phi}\dot{\theta}\sin\phi\sin\theta$$

$$= r^2\dot{\phi}^2 + \frac{1}{4}l^2\dot{\theta}^2 - rl\dot{\phi}\dot{\theta}\left(\cos\phi\cos\theta + \sin\phi\sin\theta\right)$$

$$= r^2\dot{\phi}^2 + \frac{1}{4}l^2\dot{\theta}^2 - rl\dot{\phi}\dot{\theta}\cos(\phi-\theta)$$

$$T_{cm} = \frac{1}{2}m\left(r^2\dot{\phi}^2 + \frac{1}{4}l^2\dot{\theta}^2 - rl\dot{\phi}\dot{\theta}\cos(\phi-\theta)\right)$$

$$T_{rot} = \frac{1}{2}I\dot{\theta}^2 = \frac{1}{2}\left(\frac{1}{12}ml^2\right)\dot{\theta}^2 = \frac{1}{24}ml^2\dot{\theta}^2$$

$$V = mgh = mg\left(\hat{y}\cdot\vec{r}_{cm}\right) = mg\left(-r\cos\phi + \frac{l}{2}\cos\theta\right)$$

$$\mathcal{L} = T - V$$

$$\mathcal{L} = \frac{1}{2}m\left(r^2\dot{\phi}^2 + \frac{l^2}{4}\dot{\theta}^2 - rl\dot{\phi}\dot{\theta}\cos(\phi-\theta)\right) + \frac{1}{24}ml^2\dot{\theta}^2 + mg\left(r\cos\phi - \frac{l}{2}\cos\theta\right) = 0$$

$$= \frac{1}{2}\left(r^2\dot{\phi}^2 - rl\dot{\phi}\dot{\theta}\cos(\phi-\theta)\right) + \frac{1}{6}l^2\dot{\theta}^2 + g\left(r\cos\phi - \frac{l}{2}\cos\theta\right) = 0$$

Question assigned to the following page: 2

$$\frac{d}{dt}\frac{\partial}{\partial \dot\phi}\left(\tfrac{1}{2}mr^2\dot\phi - \tfrac{1}{2}mrl\dot\theta\cos(\phi-\theta)\right) = \tfrac{1}{2}mr^2\ddot\phi - \tfrac{1}{2}mrl\ddot\theta\cos(\phi-\theta) + \tfrac{1}{2}mrl\dot\theta\sin(\phi-\theta)(\dot\phi-\dot\theta)$$

$$\frac{\partial L}{\partial \phi} = \tfrac{1}{2}mrl\dot\phi\dot\theta\sin(\phi-\theta) - mgr\sin\phi$$

$\dot\phi\dot\theta$ terms cancel

$$\tfrac{1}{2}mr^2\ddot\phi - \tfrac{1}{2}mrl\ddot\theta\cos(\phi-\theta) + \tfrac{1}{2}mrl\dot\theta\sin(\phi-\theta)(\dot\phi-\dot\theta) - \tfrac{1}{2}mrl\dot\phi\dot\theta\sin(\phi-\theta) + mgr\sin\phi = 0$$

$$2r\ddot\phi - l\ddot\theta\cos(\phi-\theta) - l\dot\theta^2\sin(\phi-\theta) + 2g\sin\phi = 0$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot\theta} = \frac{d}{dt}\left(-\tfrac{1}{2}mrl\dot\phi\cos(\phi-\theta) + \tfrac{1}{3}ml^2\dot\theta\right)$$

$$= -\tfrac{1}{2}mrl\ddot\phi\cos(\phi-\theta) + \tfrac{1}{2}mrl\dot\phi\sin(\phi-\theta)(\dot\phi-\dot\theta) + \tfrac{1}{3}ml^2\ddot\theta$$

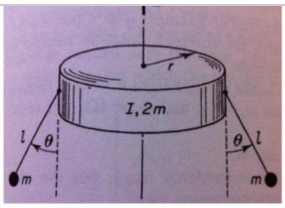
$$\frac{\partial L}{\partial \theta} = \tfrac{1}{2}mrl\dot\phi\dot\theta\sin(\phi-\theta)(-1) + \tfrac{l}{2}mg\sin\theta \qquad -\tfrac{l}{2}mg\sin\theta$$

$$-\tfrac{1}{2}mrl\ddot\phi\cos(\phi-\theta) + \tfrac{1}{2}mrl\dot\phi\sin(\phi-\theta)(\dot\phi-\dot\theta) + \tfrac{1}{3}ml^2\ddot\theta + \tfrac{1}{2}mrl\dot\phi\dot\theta\sin(\phi-\theta) = 0$$

$$-\tfrac{1}{2}mr\ddot\phi\cos(\phi-\theta) + \tfrac{1}{2}mr\dot\phi^2\sin(\phi-\theta) + \tfrac{1}{3}ml\ddot\theta - \tfrac{mg}{2}\sin\theta = 0$$

$$-3r\ddot\phi\cos(\phi-\theta) + 3r\dot\phi^2\sin(\phi-\theta) + 2l\ddot\theta - 3g\sin\theta = 0$$

Question assigned to the following page: 3



A satellite consists of a disk or radius r, mass $2m$, and moment of inertia I about the axis of symmetry. Two particles, each of mass m, are attached by rigid massless rods of length l to opposite points on the circumference. The frictionless pin joints are oriented so that the two rods always lie in a plane which contains the symmetry axis and rotates with the disk. Initially, the disk is spinning about its axis of symmetry at ω_0 rad/sec with $\theta(0) = 0$ and $\dot{\theta}(0) = 0$. For free motion, solve for the satellite rotation rate ω and also the value of $\dot{\theta}$ as functions of θ.

$I_d = \frac{1}{2}(2m)r^2 = 2mr^2 \qquad I_p = 2m(r + l\sin\theta)^2$

$H = m\omega\left(r^2 + 2(r + l\sin\theta)^2\right) = \text{const}$

$H_0 = m\omega_0\left(r^2 + 2(r)^2\right) = 3mr^2\omega_0 \qquad H = H_0$

$m\omega\left(r^2 + 2(r + l\sin\theta)^2\right) = 3mr^2\omega_0$

$2(r + l\sin\theta)^2 = \frac{3r^2\omega_0}{\omega} - r^2$

$(r + l\sin\theta)^2 = \frac{1}{2}\left(\frac{3r^2\omega_0}{\omega} - r^2\right)$

$T_d = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{1}{2}(m)r^2\right)\omega^2 = \frac{1}{4}mr^2\omega^2$ $\qquad V = 0$ no grav.

$T_{p,w} = 2\left(\frac{1}{2}m(r + l\sin\theta)^2\omega^2\right) = m\omega^2(r + l\sin\theta)^2$

$T_{p,\theta} = 2\left(\frac{1}{2}m[l\dot{\theta}]^2\right) = ml^2\dot{\theta}^2$

$L = T - V = T = \text{const.}$

$T_{rot} = \frac{1}{2}mr^2\omega^2 + m\omega^2(r + l\sin\theta)^2 + ml^2\dot{\theta}^2$

$= \frac{1}{2}mr^2\omega^2 + m\omega^2\left(\frac{1}{2}\left(\frac{3r^2\omega_0}{\omega} - r^2\right)\right) + ml^2\dot{\theta}^2$

$= \frac{1}{2}mr^2\omega^2 + \frac{3m\omega r^2\omega_0}{2\omega} - \frac{m\omega^2 r^2}{2} + ml^2\dot{\theta}^2$

$= \frac{1}{2}mr^2\omega^2 + \frac{3}{2}mr^2\omega_0 - \frac{1}{2}mr^2\omega^2 + ml^2\dot{\theta}^2$

$= \frac{3}{2}mr^2\omega_0 + ml^2\dot{\theta}^2$ $\qquad T_0 = \frac{3}{2}mr^2\omega_0^2$

$\frac{3}{2}mr^2\omega_0 + ml^2\dot{\theta}^2 = \frac{3}{2}mr^2\omega_0^2$

$\frac{3}{2}mr^2\omega_0 = \frac{3}{2}mr^2\omega_0^2 - ml^2\dot{\theta}^2$

$\frac{3}{2}mr^2$

Question assigned to the following page: 3

$$w = \frac{2}{3}\frac{\ell^2}{w_0 r^2}\dot{\theta}^2$$

$$w\left(r^2 + 2(r+\ell\sin\theta)^2\right) = 3r^2 w_0$$

$$\frac{2}{3}\frac{\ell^2}{w_0 r^2}\dot{\theta}^2\left(r^2 + 2(r+\ell\sin\theta)^2\right) = 3r^2 w_0$$

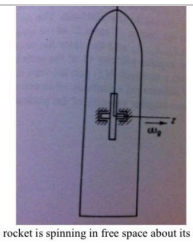
$$\dot{\theta}^2 = \frac{9r^4 w_0^2}{2\ell^2}\left(r^2 + 2(r+\ell\sin\theta)^2\right)^{-1}$$

$$\dot{\theta} = \frac{3r^2 w_0}{\ell\sqrt{2r^2 + 4(r+\ell\sin\theta)^2}}$$

$$w = \frac{2}{3}\frac{\ell^2}{w_0 r^2}\cdot\frac{9r^4 w_0^2}{2\ell^2}\left(r^2 + 2(r+\ell\sin\theta)^2\right)^{-1}$$

$$w = \frac{3r^2 w_0}{r^2 + 2(r+\ell\sin\theta)^2}$$

Question assigned to the following page: 4.1



An axially symmetric rocket is spinning in free space about its axis of symmetry with an angular velocity $\omega_r = 10$ rad/sec. Freely suspended at the mass center of the rocket by massless gimbals is a gyro rotor which is rotating at an absolute rate $\omega_g = 10^4$ rad/sec about its own axis of symmetry which is perpendicular to that of the rocket. Suddenly the gimbals are clamped and the rotor motion relative to the rocket is stopped. The gyro rotor has moments of inertia $I_a = 0.1$ kg·m^2 and $I_t = 0.05$ kg·m^2. The rocket, exclusive of the rotor, has moments of inertia $I_a = 10^3$ kg·m^2 and $I_t = 10^4$ kg·m^2. Solve for the following:

a.) The x, y, z components of the angular impulse applied to the rocket by the gyro at the instant the rotor is stopped

b.) The maximum angular deviation of the axis of symmetry of the rocket from its initial position. Note that approximate axial symmetry is maintained after the gimbals are clamped.

c.) Find the time required for a complete precession cycle of the rocket.

No external force so $\Delta H = 0$

when rotor is clamped, it gets rocket ω_r

$I_{g,a} = 0.1$ kg·m^2 $\quad I_{r,a} = 10^3$ kg·m^2

$I_{g,t} = 0.05$ kg·m^2 $\quad I_{r,t} = 10^4$ kg·m^2

$\omega_g = 10^4 \frac{1}{s}$ $\quad \omega_r = 10 \frac{1}{s}$

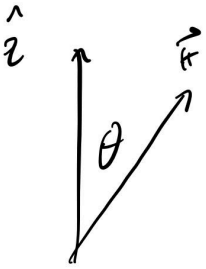
$H_{g1} = I_{g,a}\,\omega_g\,\hat{z}$ $\quad H_{g2} = I_{g,t}\,\omega_r\,\hat{x} + I_{g,a}\,\omega_g\,\hat{z}$

$\Delta H_g = H_{g2} - H_{g1} = I_{g,t}\,\omega_r\,\hat{x}$

$\Delta H_r = -0.5\,\hat{x}$ N·s

$H_1 = H_{r1} + H_{g1} = I_{r,a}\,\omega_r\,\hat{x} + I_{g,a}\,\omega_g\,\hat{z} = (10^3 \text{ kg·m}^2)(10\tfrac{1}{s})\hat{x} + (0.1 \text{ kg·m}^2)(10^4\tfrac{1}{s})\hat{z}$

$= 10^3\hat{x} + 10^4\hat{z}$ N·s $= \vec{H}_0$



First spins about $\vec{\omega}_1$ then when clamped turns into 1 rigid body in the axial direction and precesses about $\vec{H}$

$\tan\theta = \frac{H_x}{H_z} = \frac{10^3}{10^4} = 0.1$ $\quad \theta = 5.71°$

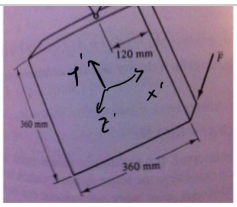
$\angle_{tot} = 11.42°$

$\dot{\psi} = \frac{\|H\|}{I_t} = \frac{\sqrt{(10^3)^2 + (10^4)^2}}{I_{g,a} + I_{r,t}} = \frac{10044.88}{10040.1} = 1.005 \frac{rad}{s}$

$T = \frac{2\pi}{\dot{\psi}}$

$T = 6.25$ s

Question assigned to the following page: 5.1

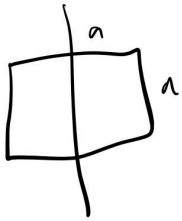


A 10-kg square plate suspended by a ball-and-socket joint is at rest when it is struck by a hammer. The impulse force $\vec{F}$ generated by the hammer is normal to the surface of the plate, and its average value during the 4-ms interval that it acts is 5,000 N. Find:

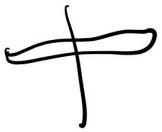
a.) The angular velocity of the plate at the instant following the impact.

b.) The average reaction at the support.

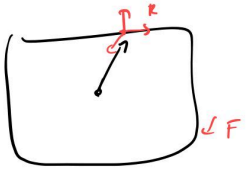
$I_t = \frac{1}{12} m a^2 = \frac{1}{12}(10kg)(.36m)^2 = 0.108\ kg\cdot m^2$



$I_a = \frac{1}{6} m a^2 = \frac{1}{6}(10kg)(.36m)^2 = .216\ kg\cdot m^2$



$\{I\}_B = \begin{bmatrix} I_{x'} & & \\ & I_{y'} & \\ & & I_{z'} \end{bmatrix} = \begin{bmatrix} 0.108 & & \\ & 0.108 & \\ & & 0.216 \end{bmatrix} kg\cdot m^2$



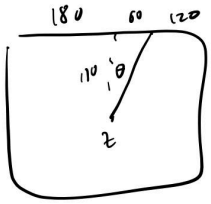
$\vec{r}_R = .06\,\hat{x}' + .18\,\hat{y}'\ m$

$\{I\}_R = \{I\} + m\left((\vec{R}\cdot\vec{R})I_3 - \vec{R}\otimes\vec{R}\right)$

$= \begin{bmatrix} 0.108 & & \\ & 0.108 & \\ & & 0.216 \end{bmatrix} + (10kg)\left((.06^2+.18^2)I_3 - \begin{bmatrix} .06^2 & .0108 & 0 \\ .0108 & .18^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}\right)$

$\{I\} = \begin{pmatrix} 0.432 & -0.108 & 0 \\ -0.108 & 0.144 & 0 \\ 0 & 0 & 0.576 \end{pmatrix}$

$M = \vec{r}\times\vec{F} = \begin{bmatrix} .18 \\ -.36 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 5000\,N \end{bmatrix} = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ .18 & -.36 & 0 \\ 0 & 0 & 5000 \end{bmatrix} = \begin{bmatrix} -1800 \\ -900 \\ 0 \end{bmatrix} N\cdot m$



$M = \frac{dH}{dt} = \frac{\Delta H}{\Delta t} \qquad \Delta H = M\Delta t = I\omega_2 - I\omega_1 = I\omega$

$\vec{\omega}_B = I^{-1} M \Delta t = \begin{bmatrix} 0.108 & & \\ & 0.108 & \\ & & 0.216 \end{bmatrix} \begin{bmatrix} -1800 \\ -900 \\ 0 \end{bmatrix} (4E\text{-}3s) \qquad \vec{\omega}_B = \begin{bmatrix} -28.205 \\ -46.154 \\ 0 \end{bmatrix} \frac{1}{s}$

$\theta = \tan^{-1}\left(\frac{60}{180}\right) = 18.435^\circ$ but -18.435° to get $B \to I$

$\vec{\omega}_I = R_{B\to I}\,\vec{\omega}_B = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \vec{\omega}_B = \begin{bmatrix} 0.9487 & -0.0166 & 0 \\ 0.0166 & 0.9487 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -28.205 \\ -46.154 \\ 0 \end{bmatrix}$

$$\vec{\omega} = \begin{bmatrix} -12.16 \\ -52.7 \\ 0 \end{bmatrix}_{\frac{1}{s}}$$

$$V_{com}|_B = \vec{\omega} \times \vec{r}_{B\to com} = \begin{bmatrix} -28.205 \\ -46.154 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0.06 \\ 0.18 \\ 0 \end{bmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -28.\ldots & -46.\ldots & 0 \\ 0.06 & 0.18 & 0 \end{vmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2.3077 \end{bmatrix} \frac{m}{s}$$

$$p_{com} = \vec{p}_{applied} + \vec{p}_{pivot} = m\vec{v}_{com} \qquad p_{pivot} = F_{piv.}\,\Delta t$$

$$F_{pivot} = \frac{\vec{p}_{appl} - m\vec{v}_{com}}{\Delta t} = \frac{1}{4\text{E-}3s}\left(\begin{bmatrix} 0 \\ 0 \\ 5000 \end{bmatrix} (4\text{E-}3) - (10kg) \begin{bmatrix} 0 \\ 0 \\ 2.3077 \end{bmatrix} \right)$$

$$\vec{F}_{pivot} = \begin{bmatrix} 0 \\ 0 \\ -769.2 \end{bmatrix} N$$

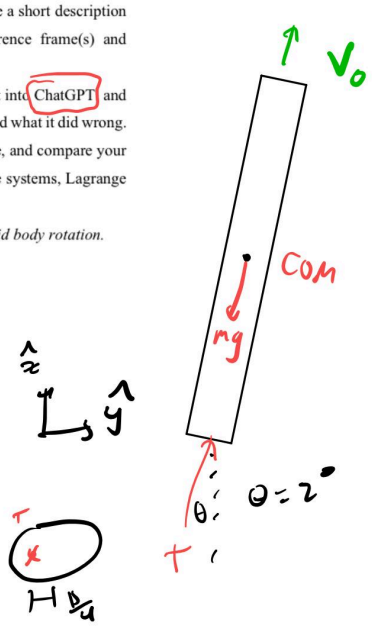
Question assigned to the following page: 6.1

Force Balance OR Lagrange. Don't forget to draw your FBD and provide a short description of what you are trying to solve. Also please label your reference frame(s) and coordinate system(s) on your drawing.

Also, as an exercise, copy and paste your creative problem prompt into ChatGPT and share the response. Make some comments about what it did right and what it did wrong.

+5pts extra credit: Solve using both Force Balance AND Lagrange, and compare your answers. If they differ, try to explain why (i.e. different coordinate systems, Lagrange doesn't handle variable-mass systems well, etc.)

Hint: use Euler angles as your generalized coordinates for 3D rigid body rotation.



(not neglecting gravity)

$\vec{r}_0 = [x_0, y_0, z_0]^T$

$\vec{v}_0 = [0, 0, V_0]^T$

$x(0) = x_0 \quad y(0) = y_0$

$\dot{x}(0) = 0 \quad \dot{y}(0) = V_0$

$w(0) = 0 \quad \theta(0) = 2°$

with rocket mass M diameter D length L $L >> D$

off axis by $e = \frac{D}{4}$

what is the maximum altitude the rocket will reach.

Reference/Coord Frames

- Body (Cartesian)
- Inertial (Cartesian)

No this is conserved during burn but every instant after burn

$$I = \frac{1}{4}M\left(\frac{D}{2}\right)^2 + \frac{1}{12}ML^2 \simeq \frac{1}{12}ML^2 \quad (L >> D)$$

$$\Sigma M = I\alpha = Te \qquad \alpha = \frac{Te}{I} = \frac{12\,Te}{ML^2}$$ before burnout

$$\Sigma F_z = T\cos\theta - Mg = Ma \qquad a = \frac{T}{M}\cos\theta - g$$

assuming small θ since $\theta_0 = 2°$ and that α is small

$$a = \frac{T}{M}\left(1 - \frac{1}{2}\theta^2\right) - g = \left(\frac{T}{M} - g\right) - \frac{1}{2}\frac{T}{M}\theta^2$$

$$\theta = \theta_0 + \frac{1}{2}\alpha t^2 \rightarrow \theta^2 = \theta_0^2 + \theta_0\alpha t^2 + \frac{1}{4}\alpha^2 t^4$$

$$\therefore \quad a = \left(\frac{T}{M} - g\right) - \frac{1}{2}\frac{T}{M}\left(\theta_0 + \theta_0\alpha t^2 + \frac{1}{4}\alpha^2 t^4\right)$$

$$V_b = V_0 + a t_b = V_0\cos\theta_0 + \left\{\left(\frac{T}{M} - g\right) - \frac{1}{2}\frac{T}{M}\left(\theta_0 + \theta_0\alpha t^2 + \frac{1}{4}\alpha^2 t^4\right)\right\} t_b$$

Question assigned to the following page: 6.1

$$\dots - \frac{T}{2m}\left(\theta_0 t_b + \theta_0 \alpha t_b^3 + \frac{1}{4}\alpha^2 t_b^5\right)$$

↓ integrate

$$z_f = X_0 + V_0 \cos\theta_0 t_b + \frac{1}{2}\left(\frac{T}{m} - g\right)t_b^2 - \frac{T}{2m}\left(\frac{1}{2}\theta_0 t_b^2 + \frac{1}{4}\theta_0 \frac{12Te}{mL^2} t_b^4 + \frac{1}{24}\left(\frac{12Te}{mL^2}\right)^2 t_b^6\right)$$

$$z_b = X_0 + V_0 \cos\theta_0 t_b + \frac{1}{2}\left(\frac{T}{m} - g\right)t_b^2 - \frac{T}{2m}\left[\frac{1}{2}\theta_0 t_b^2 + \frac{3Te\theta_0}{mL^2} t_b^4 + \frac{6T^2e^2}{m^2L^4} t_b^6\right]$$

height at end of burn

$$V_f = V_b - g(t - t_b) = 0 \quad \text{at} \quad t - t_b = \frac{V_b}{g}$$

so $$z_f = z_b + V_b(t - t_b) - \frac{1}{2}g(t - t_b)^2$$

where

$$z_b = X_0 + V_0 \cos\theta_0 t_b + \frac{1}{2}\left(\frac{T}{m} - g\right)t_b^2 - \frac{T}{2m}\left[\frac{1}{2}\theta_0 t_b^2 + \frac{3Te\theta_0}{mL^2} t_b^4 + \frac{6T^2e^2}{m^2L^4} t_b^6\right]$$

Could not use conservation equations because thrust is along the body frame as it rotates, so it is not easy to find energy exerted in x and z axes. z axis is necessary because we want to find max height.

If no initial angle

$$F \cdot t_b = m\Delta V \qquad \Delta V = \frac{F \cdot t_b}{m} \qquad T_b = \frac{1}{2}m\left(V_0 + \frac{Ft_b}{m}\right)^2 \qquad T_f = 0$$

$E_i \neq E_b$ but $E_b = E_f$ so $\frac{1}{2}m\left(V_0 + \frac{Ft_b}{m}\right)^2 = mg(z - z_0)$

$$z - z_0 = \frac{1}{2g}\left(V_0 + \frac{Ft_b}{m}\right)^2$$

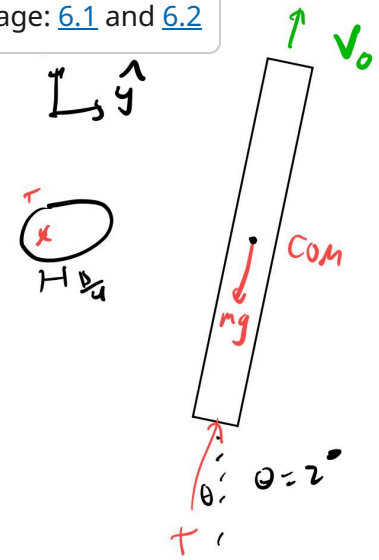
$$z = z_0 + \frac{1}{2g}\left(V_0 + \frac{Ft_b}{m}\right)^2$$

Questions assigned to the following page: 6.1 and 6.2

$x(0)=x_0 \quad y(0)=y_0$

$\dot{x}(0)=0 \quad \dot{y}(0)=V_0$

$W(0)=0 \quad \theta(0)=2^\circ$



with rocket mass M diameter D length L $L \gg D$

off axis by $e = \frac{D}{4}$

(not neglecting gravity)

$\vec{r}_0 = [x_0, y_0, z_0]^T$

$\dot{\vec{r}}_0 = [0, 0, V_0]^T$

Nice — let's keep this tidy and practical. I'll assume the rocket is a rigid body, thrust is fixed in magnitude T and applied a transverse offset e from the center of mass (you earlier used $e = D/4$), mass m is constant during the short burn t_b, and gravity g is constant. I'll do a small-angle approximation for the attitude so you get closed-form expressions you can evaluate quickly; if the resulting angle change $\Delta\phi$ is not small the closed forms break down and you should integrate numerically.

Variables / assumptions

- m — rocket mass (constant)
- T — thrust magnitude (along vehicle axis)
- e — thrust offset from COM (lever arm). ($\to e = D/4$ if you want.)
- I — pitch moment of inertia about transverse axis (approx $I \approx \frac{1}{12} m L^2$ for a slender rod).
- t_b — burn duration.
- v_0 — initial vertical velocity (along +z).
- $\phi(t)$ — pitch angle from the vertical (small), $\phi(0) = \phi_0$ (you previously used 2° as an example → $\phi_0 \approx 0.0349$ rad).
- g — gravitational acceleration ($\approx 9.81\ \mathrm{m/s^2}$).
- small-angle approximations: $\sin\phi \approx \phi,\ \cos\phi \approx 1 - \frac{1}{2}\phi^2$.

1) Rotational motion (driving $\phi(t)$)

Torque from off-axis thrust (assumed constant during burn):

$$M = Te.$$

Rotational ODE: $I\ddot{\phi} = M$. With initial $\dot{\phi}(0) = 0$,

$$\ddot{\phi} = \alpha \equiv \frac{Te}{I} \quad \Rightarrow \quad \phi(t) = \phi_0 + \tfrac{1}{2}\alpha t^2, \quad 0 \le t \le t_b.$$

Angular velocity at burn end:

$$\omega(t_b) = \dot{\phi}(t_b) = \alpha t_b.$$

(After burn, if no other moments, $\dot{\phi}$ is constant and ϕ grows linearly in coast.)

2) Vertical acceleration while burning (time-varying because of $\phi(t)$)

Thrust vertical component is $T\cos\phi(t)$. Using $\cos\phi \approx 1 - \frac{1}{2}\phi^2$,

$$a_z(t) = \frac{T}{m}\cos\phi(t) - g \approx \left(\frac{T}{m} - g\right) - \frac{T}{2m}\phi(t)^2.$$

Substitute $\phi(t) = \phi_0 + \frac{1}{2}\alpha t^2$. For most practical starts ϕ_0 is small and the ϕ_0-terms are negligible; I'll keep ϕ_0 in the formulas below.

So

$$\phi(t)^2 = \phi_0^2 + \phi_0 \alpha t^2 + \tfrac{1}{4}\alpha^2 t^4.$$

Thus (expand)

$$a_z(t) \approx \left(\frac{T}{m} - g\right) - \frac{T}{2m}\left(\phi_0^2 + \phi_0 \alpha t^2 + \tfrac{1}{4}\alpha^2 t^4\right).$$

3) Integrate to get vertical velocity and height at end of burn

Integrate a_z over 0 to t_b. Vertical velocity at burn end v_{z1} is

$$v_{z1} = v_0 + \int_0^{t_b} a_z(t)\,dt.$$

Carrying out the integrals term-by-term gives (closed-form):

$$v_{z1} = v_0 + \left(\frac{T}{m} - g\right)t_b - \frac{T}{2m}\left(\phi_0^2 t_b + \phi_0 \alpha \tfrac{t_b^3}{3} + \tfrac{1}{4}\alpha^2 \tfrac{t_b^5}{5}\right).$$

Height at burn end z_1 (starting $z(0) = 0$) is

$$z_1 = \int_0^{t_b}\left(v_0 + \int_0^{\tau} a_z(\tau)d\tau\right)dt = v_0 t_b + \tfrac{1}{2}\left(\frac{T}{m} - g\right)t_b^2 - \frac{T}{2m}\left(\phi_0^2 \tfrac{t_b^2}{2} + \phi_0 \alpha \tfrac{t_b^4}{12} + \tfrac{1}{4}\alpha^2 \tfrac{t_b^6}{30}\right).$$

Those are exact within the small-angle Taylor expansion used ($\cos \approx 1 - \frac{1}{2}\phi^2$). If you set $\phi_0 = 0$ they simplify:

If $\phi_0 = 0$:

$$v_{z1} = v_0 + \left(\frac{T}{m} - g\right)t_b - \frac{T}{2m}\cdot\frac{\alpha^2 t_b^5}{20},$$

$$z_1 = v_0 t_b + \tfrac{1}{2}\left(\frac{T}{m} - g\right)t_b^2 - \frac{T}{2m}\cdot\frac{\alpha^2 t_b^6}{120}.$$

Recall $\alpha = \frac{Te}{I}$, so the α^2 terms scale like $(Te/I)^2$ and appear with high powers of t_b (t_b^5 and t_b^6) — meaning rotation effects on the vertical are small for short burns or small e.

4) Maximum altitude after burn

Assuming the apex happens after burn (i.e. vertical velocity is still upward at burn end), the additional rise after burn is the usual ballistic term:

$$\Delta z_{\text{coast}} = \frac{v_{z1}^2}{2g}, \qquad z_{\max} = z_1 + \frac{v_{z1}^2}{2g}.$$

If $v_{z1} \le 0$, then the apex occurs during the burn — in that case you must find time t^* in $[0, t_b]$ where $v_z(t^*) = 0$ using the full $v_z(t)$ polynomial above (one can solve numerically or analytically for that root) and compute $z(t^*)$.

5) Horizontal (downrange) effects — quick formula (small angle)

Horizontal acceleration $a_x(t) \approx (T/m)\sin\phi(t) \approx (T/m)\phi(t)$. With $\phi(t) \approx \phi_0 + \frac{1}{2}\alpha t^2$ and ϕ_0 small,

$$v_{x1} = \int_0^{t_b} a_x(t)\,dt \approx \frac{T}{m}\left(\phi_0 t_b + \frac{\alpha t_b^3}{6}\right).$$

So rotation-driven contribution to horizontal velocity grows like t_b^3 (if ϕ_0 negligible) and depends on $\alpha = Te/I$. Downrange distance after ballistic coast uses the impact time from the quadratic in z (same as earlier):

$$t_{\text{impact}} = \frac{v_{z1} + \sqrt{v_{z1}^2 + 2gz_1}}{g}, \qquad x_{\text{landing}} \approx x_1 + v_{x1}\,t_{\text{impact}},$$

with $x_1 \approx \frac{T}{m}\left(\phi_0 \frac{t_b^2}{2} + \frac{\alpha t_b^4}{24}\right)$ (integral of v_x during burn).

6) Substitute $I \approx \frac{1}{12}mL^2$ and $e = D/4$ (engineer's quick form)

If you use $I \approx \frac{1}{12}mL^2$ and $e = D/4$, then

$$\alpha = \frac{Te}{I} \approx \frac{12Te}{mL^2} = \frac{12T(D/4)}{mL^2} = \frac{3TD}{mL^2}.$$

Plug that into the α-terms above to compute the small corrections due to rotation.

7) When is rotation negligible?

Compute the leading rotation-induced correction magnitude to vertical velocity:

$$\delta v_{z,\text{rot}} \sim \frac{T}{2m}\cdot\frac{\alpha^2 t_b^5}{20} \propto \frac{T}{m}\left(\frac{Te}{I}\right)^2 t_b^5.$$

If this term is $\ll (T/m)\,t_b$ (the main thrust contribution) then rotation hardly changes the max altitude. Practically, if the angular deflection during burn $\Delta\phi \approx \frac{1}{2}\alpha t_b^2 \ll 0.1$ rad (say), you're in the safe small-angle regime and the above formulas are accurate.

8) If you want numbers / a verification

I can plug in your numbers (give me $m, T, D, L, t_b, v_0, \phi_0$ or let me use $e = D/4$ and $I = \frac{1}{12}mL^2$) and compute:

- α, $\phi(t_b)$, $\Delta\phi$
- v_{z1}, z_1 (with rotation terms), $z_{\max}$
- landing range x_{landing}.

If $\Delta\phi$ is not small (say > 0.1 rad) I'll instead integrate the exact ODEs numerically and plot the trajectory+attitude. Which would you prefer — plug numbers now, or should I run a numerical integration and plot for sample values?